CURRICULUM VITAE

# Rahul Rawat

Master Thesis, Data Science and Machine Learning in Injection Molding

## PERSONAL DETAILS


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| **GitHub** | github.com/rahulrawat20022002 |
| **Portfolio** | rah-portfolio.pages.dev |
| **Date of birth** | 20 February 2002 |
| **Nationality** | Indian, student visa with valid work permit |
| **Availability** | Werkstudent 20 hours per week immediately, full time from April 2027 |


## PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on delivery of machine learning pipelines on real sensor and process data, honest leakage aware evaluation, and end to end analytics for technical stakeholders. At eRay GmbH I built a recursive time series pipeline forecasting four water quality indicators, benchmarking six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and shipped it with strict anti leakage rules and asymmetric **80 percent** prediction intervals. Proficient in Python, pandas, scikit learn, and comfortable across the wider ML stack, I am the right fit for the Master's Thesis in Data Science and Machine Learning in Injection Molding at Freudenberg Technology Innovation in Weinheim.

## PROFESSIONAL EXPERIENCE

**Oct 2025 to Mar 2026**
Heidelberg

### eRay GmbH

Data Scientist

- During a six month eRay GmbH and SRH Heidelberg collaboration to forecast water quality on a German lake, tasked with covering four indicators over rolling horizons, built an end to end recursive time series pipeline for chlorophyll a, turbidity, pH, and dissolved oxygen, delivered as a production ready module the client can rerun on every new sensor drop.
- Faced with model choice ambiguity for the forecasting task and the need to convey uncertainty to non technical stakeholders, benchmarked six candidates head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, then landed on CatBoost multi quantile regression, which produced asymmetric **80 percent** prediction intervals that gave the client decision support under uncertainty.
- Concerned that naive time series splits would inflate accuracy, tasked with making the evaluation defensible, enforced strict anti leakage rules across the pipeline, which surfaced the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- With winter readings missing and tree based models flatlining during recursive prediction, needing realistic seasonal shape in the downstream forecasts, reconstructed missing values with MICE imputation and engineered a synthetic winter decay forecast canvas, which restored believable winter behaviour without introducing bias into the training window.
- To prevent bad data cascading downstream through the recursive forecaster, tasked with hardening the run loop, wrapped the pipeline in an orchestrator with gate checks and velocity and ecological bounds, so a failed imputation now halts the run rather than corrupting weeks of downstream predictions.

**Technologies:** *Python, CatBoost, Prophet, scikit learn, MICE*

**Aug 2023 to Aug 2024**
India

### SS Engineers and Contractors
Junior Associate Software Developer

- With SS Engineers and Contractors running internal Data Dashboards, Analytics platforms, and Employee Portals used across the company, tasked with building and maintaining the front end features that day to day teams depended on, contributed React UI components across all three internal products, which stayed in daily use by internal teams across the company throughout the year in role.
- With a client relying on a legacy AngularJS app that had to sit inside an existing module federation setup alongside newer React micro frontends, tasked with porting the client's screens across without breaking the running product, ported around 8 routes from AngularJS to React inside the module federation shell over **4 months** of incremental releases, shipped with no production incidents during rollout.

**Technologies:** *React, module federation, Playwright, AngularJS, HTML5, CSS3*

**Feb 2023 to Aug 2023**
India

### SS Engineers and Contractors
Front End Developer Intern

- During a six month internship at SS Engineers and Contractors, tasked with learning the codebase and then taking on small UI work across the internal Data Dashboards and Employee Portals under senior review, paired closely with senior developers to walk the codebase and then shipped focused UI components including charts, filters, and profile pages, iterating each one on code review feedback until it landed.

**Technologies:** *React, HTML5, CSS3, Git, code review workflow*

## PERSONAL PROJECTS

**2025**

### Movie Analytics and ML Pipeline on GCP

- For a personal cloud data project on movie analytics that needed to run unattended on GCP, tasked with the ingestion and processing layer, built an end to end batch pipeline that pulls movie data from a public API into a GCS data lake and processes it through Bronze to Silver to Gold medallion architecture in BigQuery on Cloud Run, running on a fully automated Cloud Scheduler trigger with no manual intervention required.
- With raw ingested data being unfit for direct analytics, tasked with hardening the Silver layer, applied schema enforcement, safe type casting, deduplication via window functions, and genre normalisation into a relational model, which held clean referential integrity across every downstream Gold table.
- Because a classifier trained on post release signals would leak the answer, tasked with a leakage free hit prediction, trained a BigQuery ML classifier that predicts whether a film will be a hit before release, deliberately splitting features into two tables so only pre release signals feed the model, and confirmed leakage free evaluation on the split.

**Technologies:** *GCP BigQuery, Cloud Run, GCS, Cloud Scheduler, BigQuery ML, Python, SQL, Looker Studio*

**2025**

### Real Time Flight Tracking Data Pipeline

- For a Data Engineering module at SRH Heidelberg needing a real time joined view over live aircraft above Germany, tasked with the collection and enrichment layer, built Python collectors that poll the OpenSky Network API every **30 seconds** and PySpark cleaning on Google Cloud that joins against airport, aircraft, and weather data across four sources, producing a clean joined table covering more than **128 thousand** records.
- With raw collector output not being usable for analysis and each aircraft needing a nearest airport label, tasked with the modelling layer, shaped the data into analysis ready tables with dbt and computed each aircraft's nearest airport with PySpark for heavy lift, which produced consistent nearest airport labels across the historical dataset.
- Because manual reruns would kill the real time promise, tasked with automating refresh, orchestrated the whole system with Apache Airflow on GCS backed storage and Dataproc compute so that batch and real time layers refresh automatically every **15 minutes** without operator intervention.

**Technologies:** *Python, PySpark, BigQuery, dbt, Apache Airflow, GCP Dataproc and GCS, Tableau with TabPy*

## EDUCATION

**Apr 2025 to Present**
Heidelberg

### M.Sc. Data Science and Analytics
SRH University of Applied Sciences Heidelberg, GPA 1.9

**2019 to 2023**
Greater Noida, India

### Bachelor of Technology in Computer Science
GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

## RESEARCH AND THESIS

**2022 to 2023**
Greater Noida, India

### Bachelor Thesis, Diabetes Prediction Using Machine Learning
GL Bajaj Institute of Technology and Management, IEEE style paper

- For a Bachelor thesis on diabetes prediction with a small clinical dataset of **768 patients**, tasked with building a defensible model comparison the examiners could audit, built a full end to end machine learning pipeline comparing six classifiers with **10 fold** cross validation and per model confusion matrices, delivering a model comparison that stood up in the thesis defence.
- Spotting biologically impossible zero values in the source data that the original authors had overlooked, tasked with restoring data integrity before any model fit, applied IQR based outlier removal and proper imputation, which lifted the dataset from silently broken to a clean training input for every downstream model.
- With a **65 to 35** class imbalance that made accuracy a misleading headline metric, tasked with choosing an evaluation that would not hide errors on the minority class, moved the headline metric from accuracy to ROC AUC, which exposed the real error patterns the accuracy score had been masking and gave the thesis an honest performance comparison.
- With the results needing to be publishable in substance rather than just submissible, tasked with writing them up formally, produced an IEEE style paper including an honest limitations section and what to do differently in a follow up study, which the supervisor accepted as publishable in substance.

**Technologies:** *Python, scikit learn, Pandas, Seaborn*

## CERTIFICATIONS

- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- SAS Certified Specialist: Visual Business Analytics Using SAS Viya, issued May 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

## ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

## LANGUAGES

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| **English** | Fluent, written and spoken |
| **German** | B1 in progress toward B2 |
| **Hindi** | Native |